

Flutter Interview Questions & Answers

Q. What is Flutter?

Answer: Flutter is an open-source UI toolkit developed by Google that allows developers to build cross-platform applications for Android, iOS, Web, and Desktop using a single codebase.

Q. Why is Flutter different from other frameworks?

Answer: Flutter uses the Skia C++ rendering engine, which allows it to render UI directly without relying on native components, resulting in high performance.

Q. What is Dart and why does Flutter use it?

Answer: Dart is a programming language developed by Google. Flutter uses Dart because it supports object-oriented programming, JIT and AOT compilation, and helps in writing fast, maintainable code.

Q. What is a Widget in Flutter?

Answer: A widget is the basic building block of the Flutter UI. Everything in Flutter is a widget.

Q. Difference between StatelessWidget and StatefulWidget?

Answer: StatelessWidget does not change its state during runtime, while StatefulWidget can change its state dynamically.

Q. Explain Widget lifecycle.

Answer: The widget lifecycle includes createState, initState, didChangeDependencies, build, didUpdateWidget, deactivate, and dispose.

Q. What are Keys in Flutter?

Answer: Keys are used to uniquely identify widgets and preserve their state during rebuilds.

Q. Difference between GlobalKey and LocalKey?

Answer: GlobalKey works within the same widget tree, while LocalKey allows accessing widget state globally.

Q. What is Hot Reload?

Answer: Hot Reload allows developers to see UI changes instantly without restarting the app.

Q. What is Hot Restart?

Answer: Hot Restart rebuilds the entire application and resets the app state.

Q. What are Flutter build modes?

Answer: Flutter has three build modes: Debug, Release, and Profile.

Q. Difference between Container and SizedBox?

Answer: Container supports decoration, margin, padding, while SizedBox is mainly used to create space between widgets.

Q. How does Flutter handle responsive design?

Answer: Flutter uses MediaQuery, Expanded, Flex, and AspectRatio to make apps responsive.

Q. What is Navigator in Flutter?

Answer: Navigator manages routing and navigation between screens using push, pop, and pushReplacement.

Q. How do you implement animations in Flutter?

Answer: Animations can be implemented using Tween animations or third-party libraries.

Q. What are state management solutions in Flutter?

Answer: Common state management solutions include Provider, GetX, and Bloc.

Q. What is setState()?

Answer: setState() is used to rebuild the UI when the internal state of a widget changes.

Q. What is InheritedWidget?

Answer: InheritedWidget allows data sharing down the widget tree efficiently.

Q. How do you make HTTP requests in Flutter?

Answer: HTTP requests are made using the http or Dio package.

Q. How do you implement local storage in Flutter?

Answer: Local storage is implemented using SharedPreferences, SQFLite, or Hive.

Q. How do you encrypt sensitive data in Flutter?

Answer: Sensitive data can be encrypted using flutter_secure_storage, crypto, or AES libraries.

Q. What types of testing are available in Flutter?

Answer: Flutter supports Unit testing, Widget testing, and Integration testing.

Q. How do you debug performance issues?

Answer: Performance issues are debugged using Flutter profile mode and DevTools.

Q. What is Dependency Injection in Flutter?

Answer: Dependency Injection is implemented using packages like GetIt.

Q. How does Flutter communicate with native code?

Answer: Flutter communicates with native Android and iOS code using MethodChannels.

Q. How do you implement push notifications?

Answer: Push notifications are implemented using Firebase Cloud Messaging (FCM).

Q. How do you prepare a Flutter app for production?

Answer: For Android, generate an App Bundle and upload to Play Store. For iOS, archive the app and upload via TestFlight.